IMD3002 – 3D COMPUTER GRAPHICS

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L14 – Lighting: Technical Methods

Overview

- This lecture covers the technical properties of various types of lights used in 3D graphics; including Image Based Lighting (IBL)

Lights

- Introduction
  - Depending on the system being used, there are a variety of different light types – most systems support approximately the following lights:
    - Ambient Light
    - Directional/Distant Light
    - Point/Omni Lights
    - Spot Lights
    - Area Light
    - Volume Light

Lights

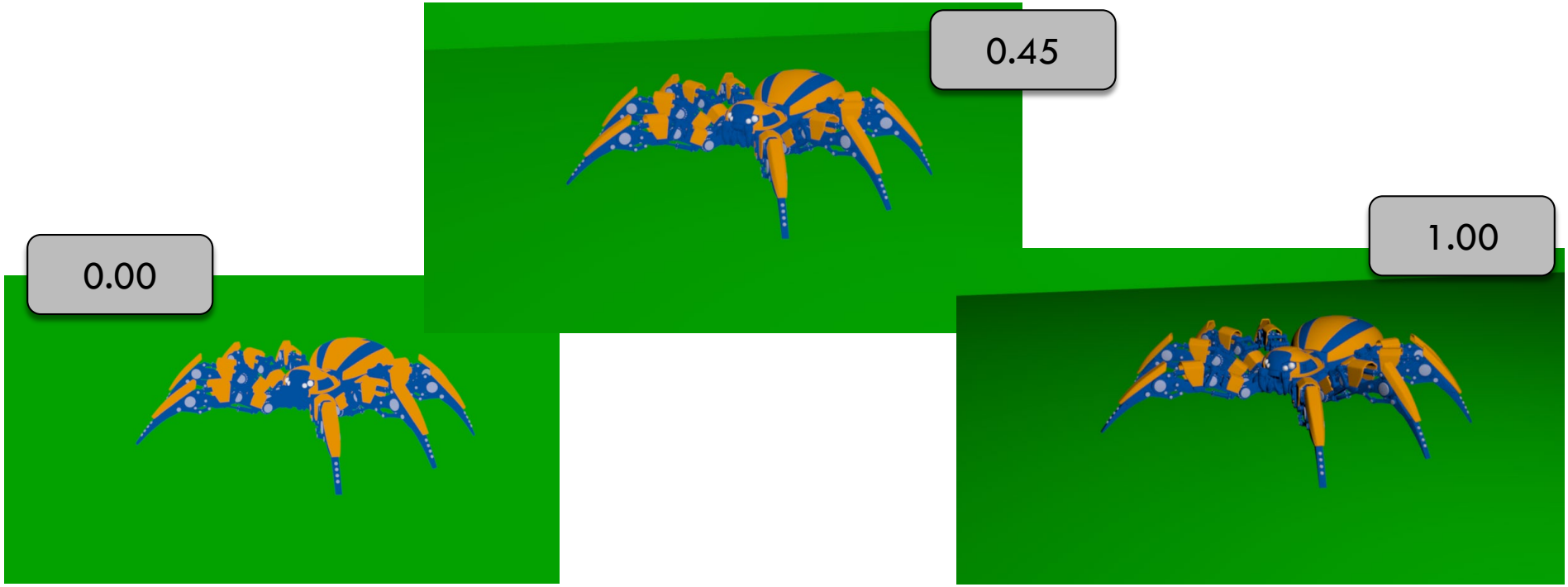
- Ambient Lights
  - Ambient lights are used to uniformly add a brightness value to all surfaces in a scene
  - It softens the effect of other lights and is used to compensate for the lack of light because there are no light bounces, or there are only a few light bounces
  - Generally ambient lights have no position or direction
    - Without position, light decay is irrelevant in this case
  - The intensity and colour of the light can be adjusted

Lights

- Ambient Lights
  - Example:
    - With only an ambient light in the scene, there should be no shadows (as it has no location), also there is no shading, so everything looks very flat

Lights

- Ambient Lights
  - A feature in Maya permits setting the proportionality of omnidirectional to directional light for an ambient light
    - This is referred to as “Ambient Shade”

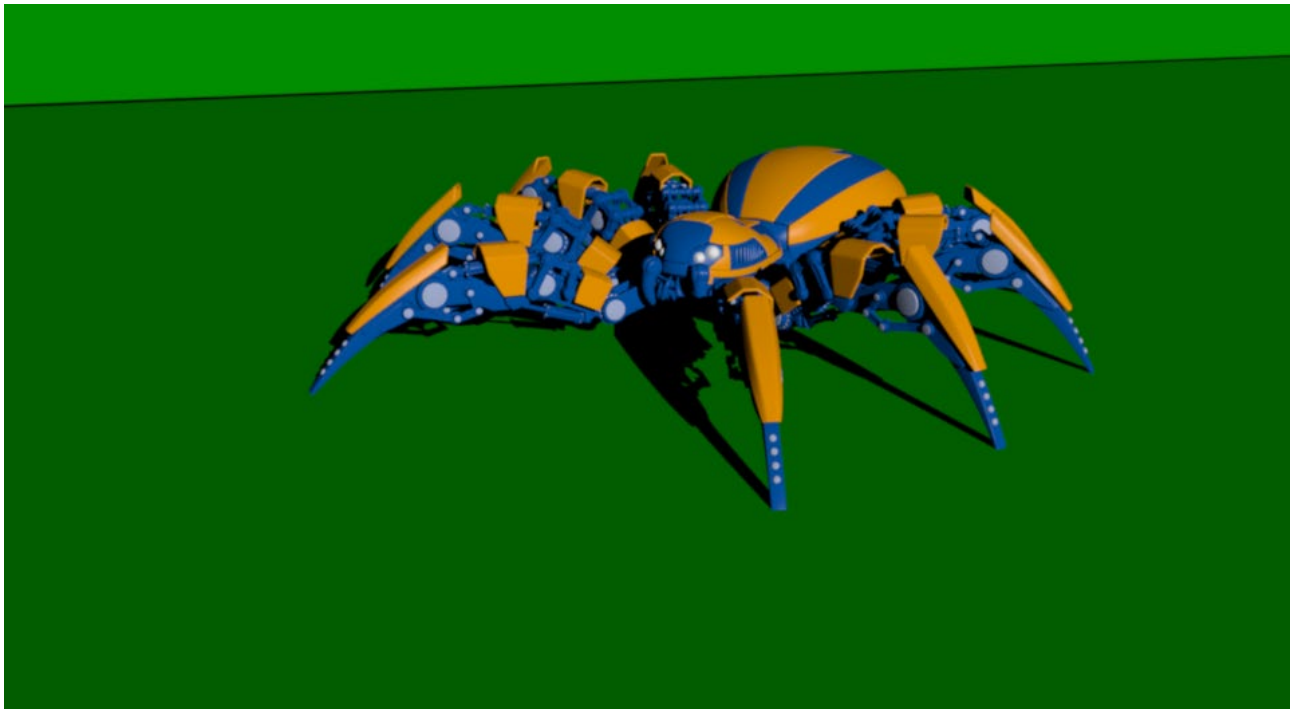


Lights

- Directional Lights
  - A directional light (or distant light) acts like the sun – it illuminates a scene uniformly in one direction
  - Objects vary in brightness according to the inclination of their surfaces to the direction of the light only
    - The (relative) position of a surface and light is irrelevant
  - The orientation of the light (set through the lights transform) is the important factor
    - Without position, light decay is irrelevant in this case
  - The intensity and colour of the light can be adjusted
  - Diffuse and specular light emission can be switch on/off

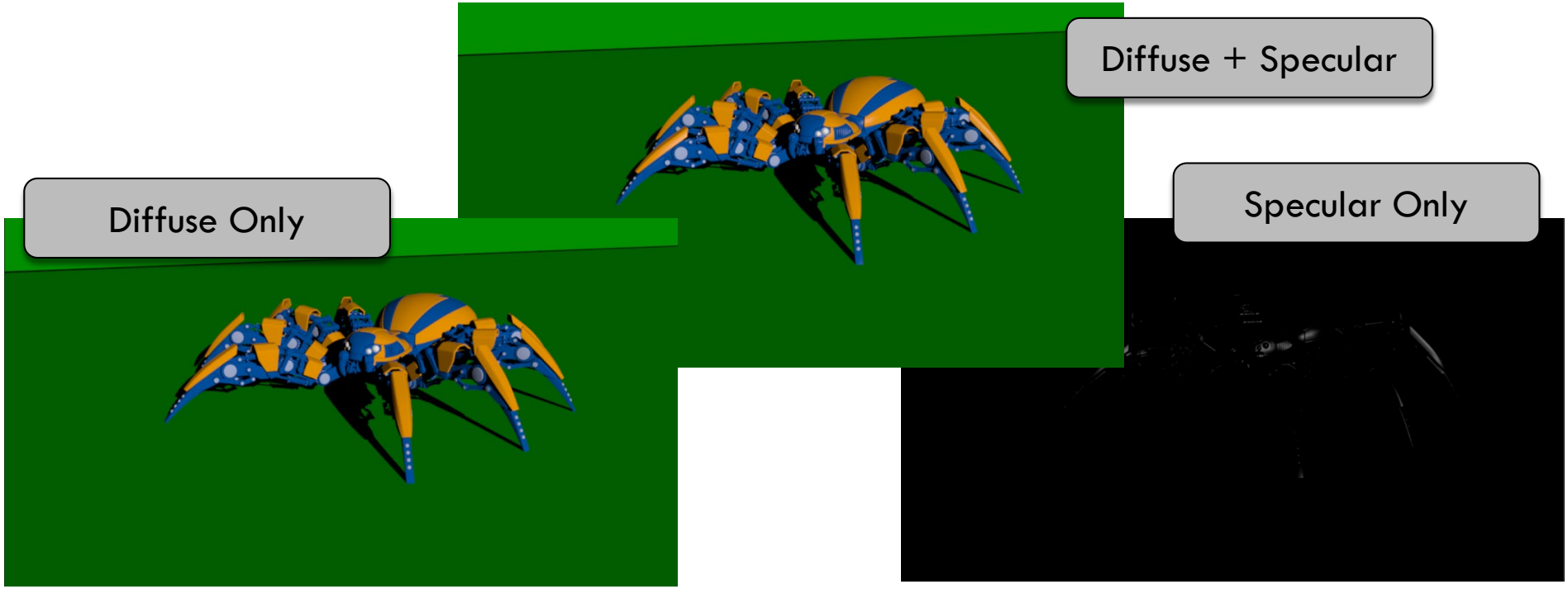
Lights

- Directional Lights
  - Example:
    - With only a directional light in the scene, there is no light decay, but shadows are clearly visible



Lights

- Directional Lights
  - Example:
    - As this type of light has direction, diffuse and specular lighting equations can be switched on/off accordingly



Lights

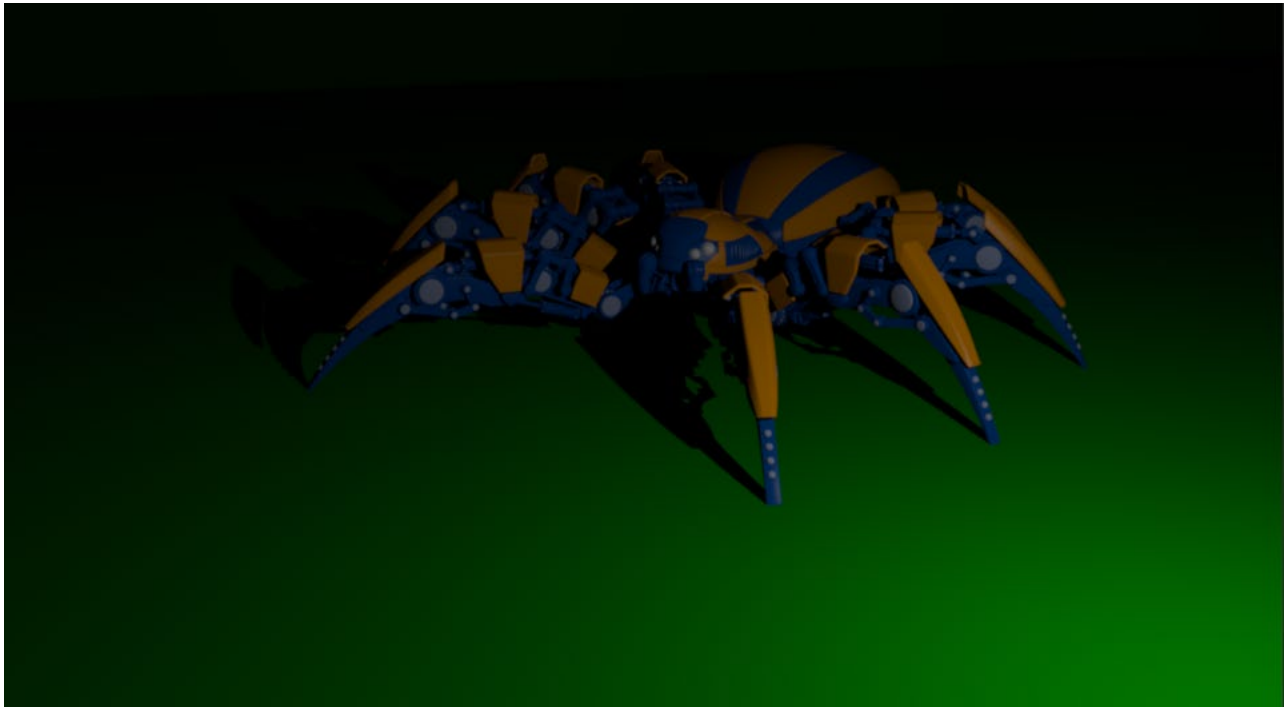
- Point Lights
  - A Point Light (or Omni Light) is like an unshielded electric light bulb; a point that source radiates light uniformly in all directions (i.e. it is Omni-directional)
  - As it has a position, but no direction, light can diminish over distance (which is referred to as decay)
    - Decay can be adjust to be:
      - Off (No Decay)
      - Linear ($1/d$)
      - Quadratic ($1/d^2$), this is how regular light behaves
      - Cubic ($1/d^3$)
  - The intensity and colour of the light can be adjusted
  - Diffuse and specular light emission can be switch on/off

Lights

- Decay
  - In real life, decay values for most types of energy (light, heat, sound, etc.) follow the quadratic decay ($1/d^2$); but as this can be quite cumbersome most graphics packages allow for other types

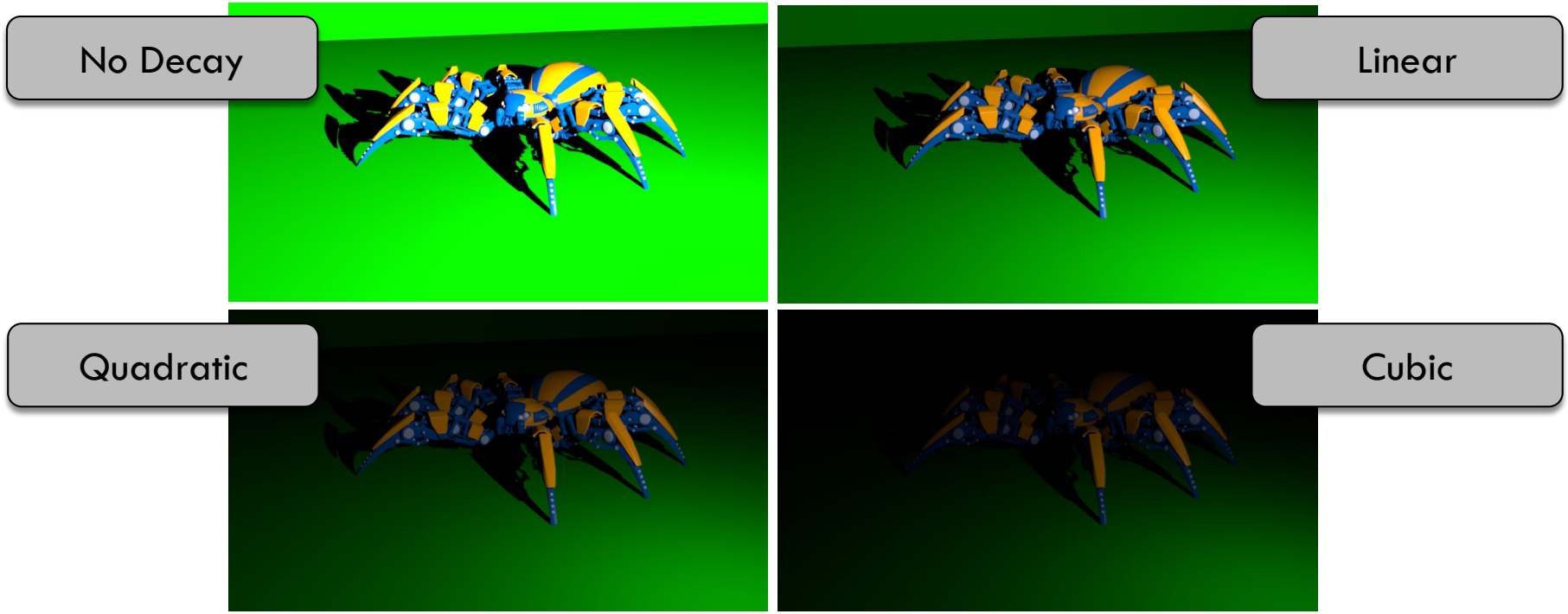
Lights

- Point Lights
  - Example:
    - As a decay value can be adjusted (in this case $1/d^2$) higher intensity values are required to illuminate an object



Lights

- Point Lights
  - Example:
    - This shows the same light intensity (6.0) with the range of decay possibilities

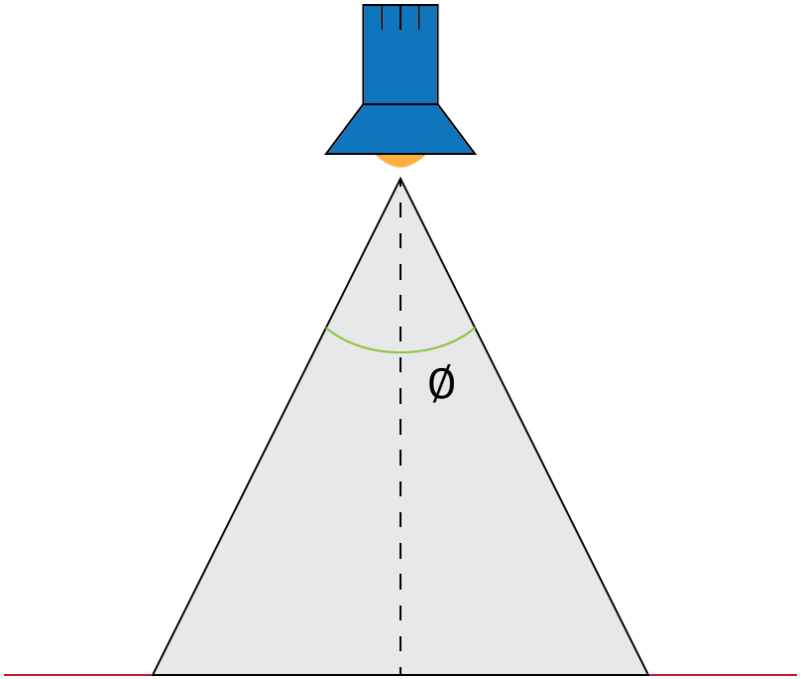


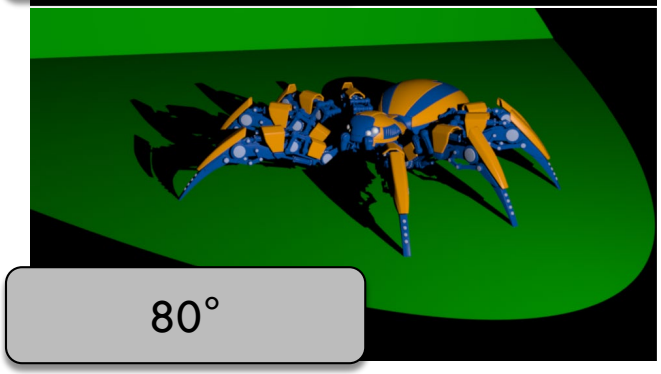
Lights

- Spotlights
  - Spotlights have a much wider range of controllable parameters compared to other light types
    - Light Colour & Intensity
    - Position & Direction (set through the node's transform)
    - Decay (None, Linear, Quadratic, Cubic)
    - Cone Angle
    - Penumbra Angle
    - Drop-off
    - Barn Doors

Lights

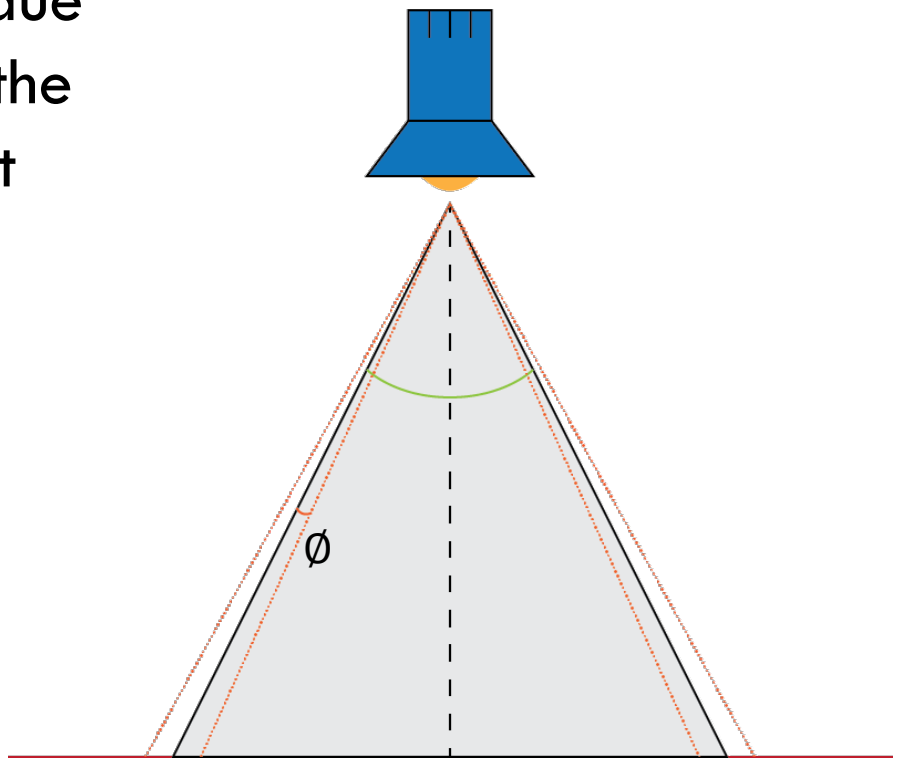
- Spot Lights
  - The Cone Angle (which specifies the actual range that the light is visible) as an arc, is specified between 0° and 180° (which is essentially flat and doesn't make a cone)





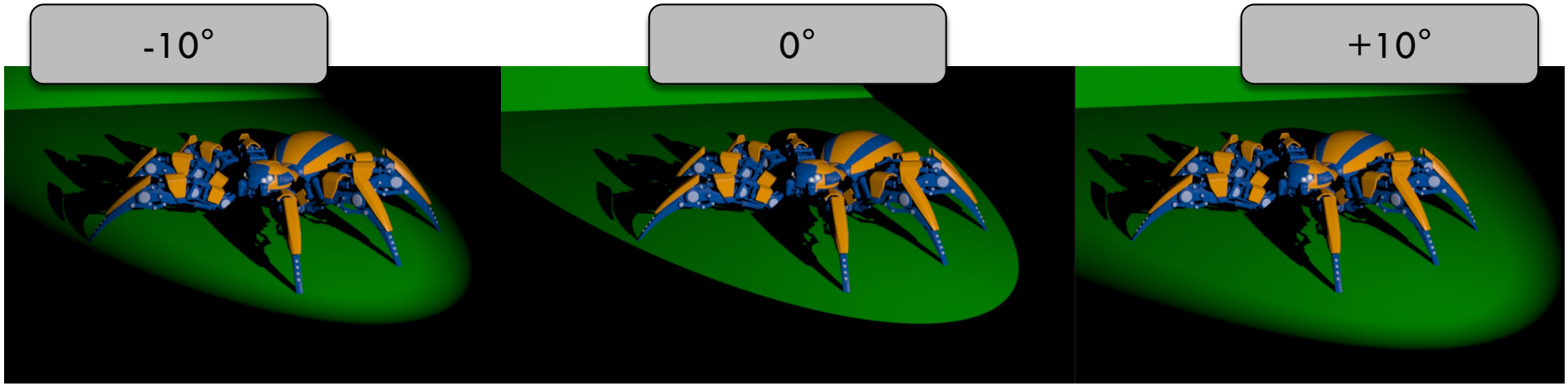
Lights

- Spot Lights
  - The Penumbra Angle defines the range (in degrees) the light will decrease from the edge, it simulates the fuzzy edges of the light cone due to light bounces around the edge and inside the light



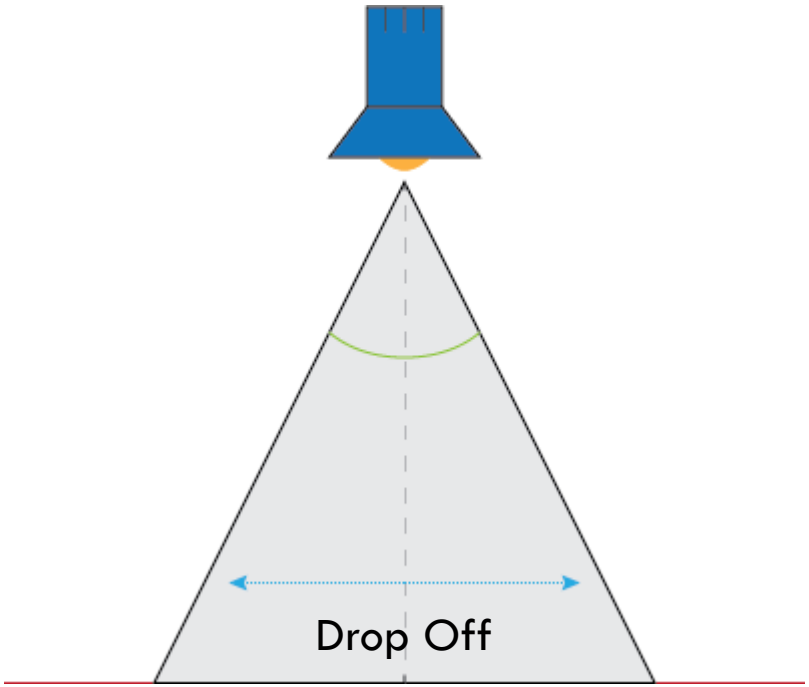
Lights

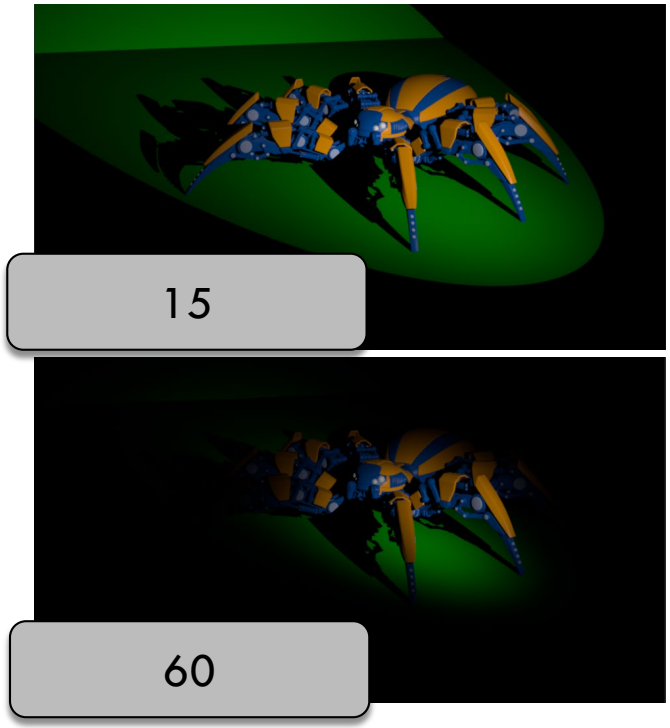
- Spot Lights
  - The Penumbra Angle is generally limited to -10° to +10°, but it can be adjusted by -179.994° to +179.994° (less or greater than the edge defined by the cone)



Lights

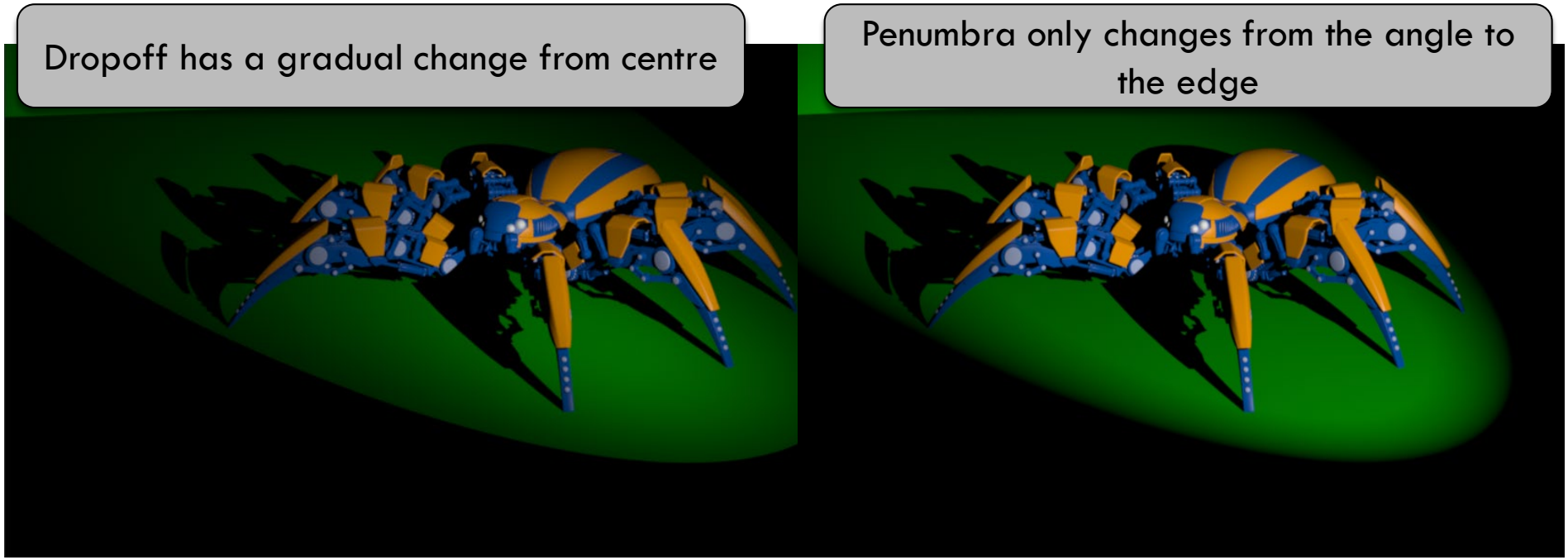
- Spot Lights
  - The Drop Off has a similar effect to the Penumbra Angle, but it's specified in a different way, it specifies how light drops off from the centre of the light





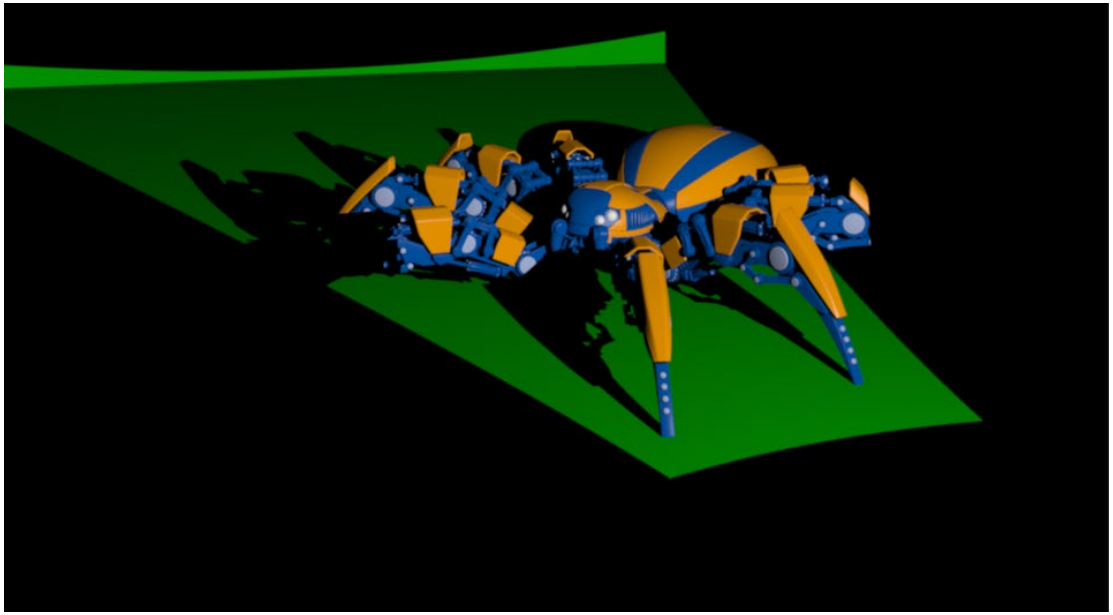
Lights

- Spot Lights
  - The difference between the Dropoff and the Penumbra Angle (apart from the unit value used) is the way the light intensity changes from the centre of the spot



Lights

- Spot Lights
  - Barn Doors work in the same way as they do on real lights; they create a linear delimitation of the light
  - These can be adjusted for all sides (top/bottom, left/right)



Lights

- Area Lights
  - An Area Light is different from previous lights as it actually has an area and is not coming from a specific point or direction; it is represented as a rectangle
  - In general, Area Lights are more representative of various types of light source
    - **Example:** sun through windows or large fluorescent tube sections in offices
  - As they are not coming from a specific direction or position, but over an area, the shadows tend to be softer when ray-traced (and appear similar to Depth Map shadows)

Lights

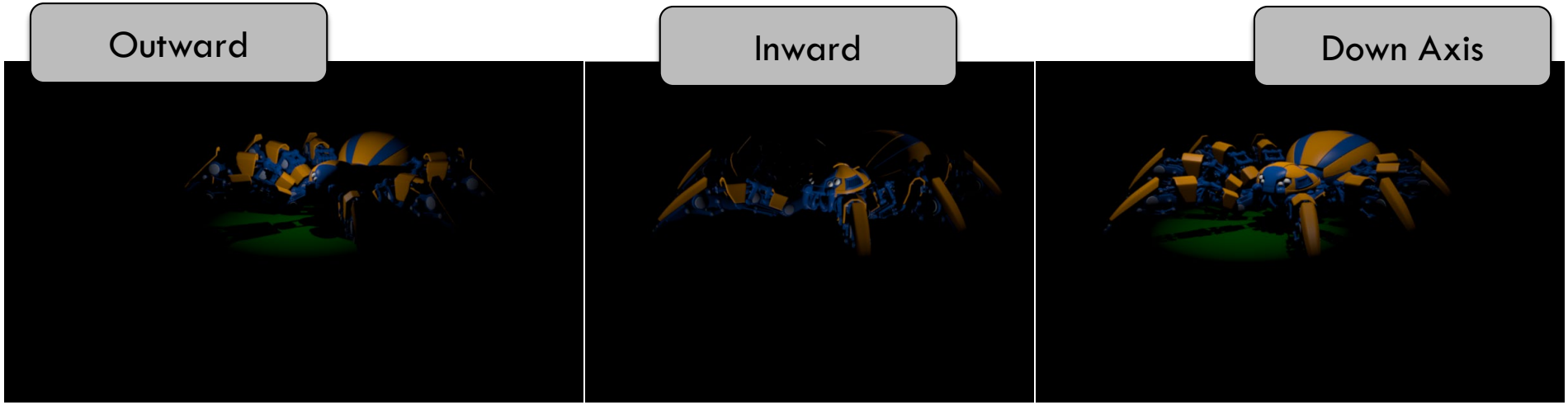
- Area Lights
  - The light calculation deals with the angle from the illuminated point and the light itself, this calculation provides decay as part of the calculation

Lights

- Volume Lights
  - Volume lights are not available on every system and can be more specific to modelling packages (rather than game engines)
  - A volume replicates the idea of a point light (with position), but provides the control of a spot light
  - The centre of the volume is the source of the light
  - Outside the boundaries of the volume there should be no light; it's contained within the boundary which is defined by the light's transform

Lights

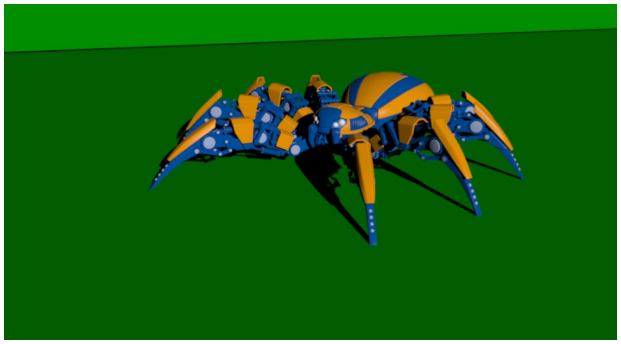
- Volume Lights
  - A Volume Light can have the Light Direction specified (with relation to the volume itself
    - Outwards – Working similar to a Point Light
    - Inwards – Edge of the volume is the light
    - Down Axis – Works similar to a Directional Light



Lights

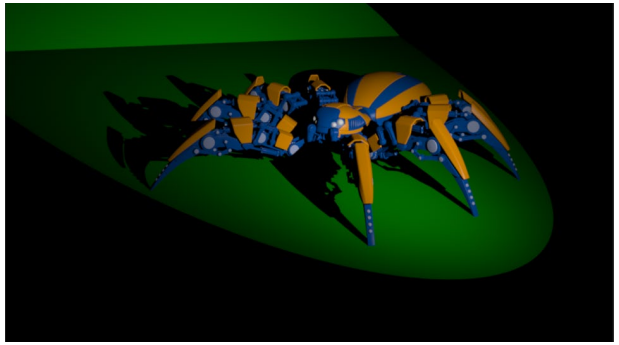
- Summary of Light Types

Ambient



Directional

Point



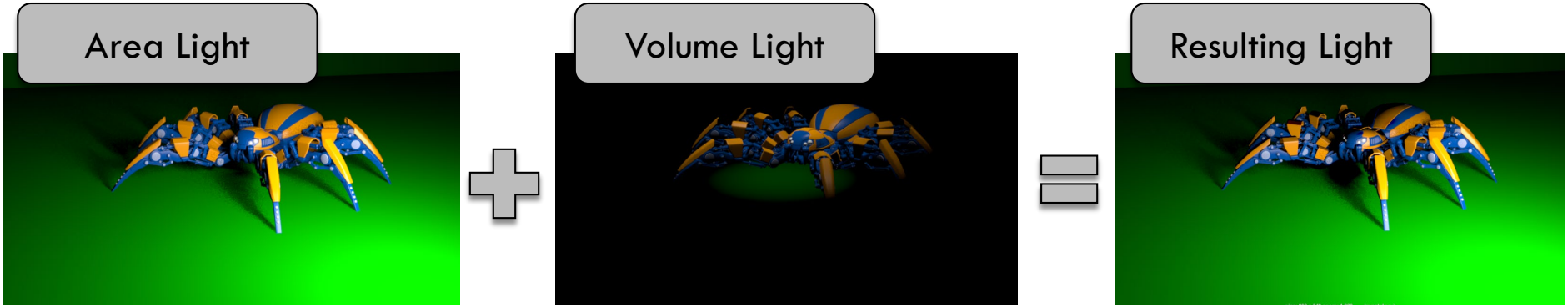
Spot

Area

Volume

Lights

- Negative Lights
  - As the lighting equation is simply a summation of all the lighting information, a negative intensity value can be used to remove light
  - In this case, if shade or shadow are required where it would be difficult specific structures, a light with a negative intensity value can be used instead



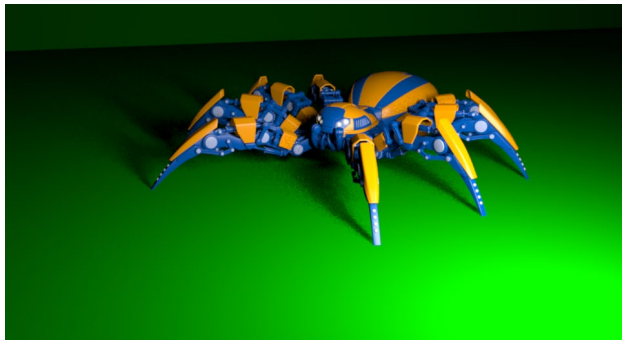
Lights

- Light Linking
  - When a light source is created, the new light source illuminates all surfaces in the scene by default
    - Assuming the light source's properties are set appropriately
  - Similarly, when a surface is created, all lights in the scene illuminate the new surface
  - To provide more control lights can be linked with surfaces so that only a specific light(s) illuminates a specific surface(s), or, alternatively, only a specific object(s) receives illumination from a specific light(s)

Lights

- Light Linking
  - Light linking can also help to render scenes more efficiently and quickly (as it reduces the lighting calculation)
  - Light linking allows very fine control over which lights affect which objects in the scene

Spot Light linked to Nothing



Spot Light linked to All

Spot Light linked to Floor

Light Colour

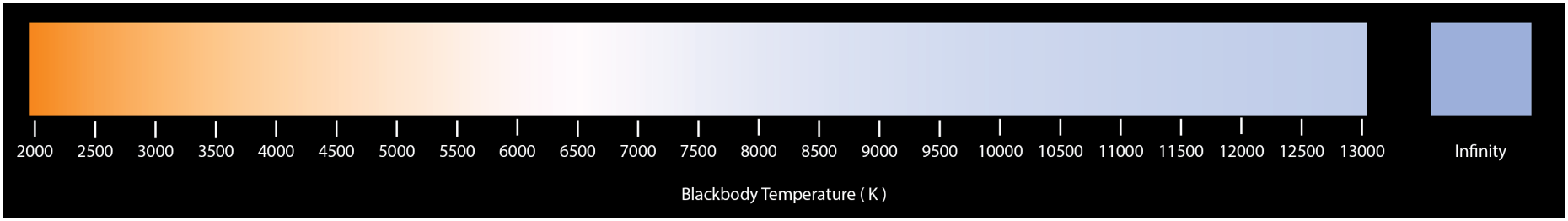
- Introduction
  - The colour temperature of a light in the physical world determines how the scene will be perceived (in terms of mood, etc)
  - Technically speaking, the colour temperature of a light source is determined by comparing its hue with a theoretical, heated, black-body radiator
  - The colour temperature of a black-body radiator is equal to its surface temperature in Kelvin (all colour temperatures are measured in Kelvin's)

Light Colour

- Introduction
  - Colour temperature in 3D graphics is also very important for realism, for two keys reasons:
    - **Mood Setting** ~ Firstly, it enables us to create specific moods; we can specify the colour temperature of a light source and achieve that specific effect
    - **Live/3D Compositing** ~ Secondly, if we need to match lights when compositing a real and a virtual scene together, then we need to know the exact colour temperature in order to replicate the same light settings in the virtual as we have in the real – otherwise the audience will be otherwise …

Light Colour

- Temperatures
  - The actual temperatures in general use are provided below
  - Colours which are generally 'acceptable' as 'white' are usually quite close to black body colours upwards of around 2500°K
    - **Note:** it is not physically possible to accurately mimic blackbody colours cooler than 1900K on a standard sRGB monitor



Light Colour

- Temperatures
  - Temperature is more important, is when we compare it to specific sources of light
    - **Example:** some common light sources

1700°K ~ Match flame
1850°K ~ Candle
2800°K ~ Incandescent lamp (Tungsten)
3350°K ~ Standard Studio light
3400°K ~ Studio lamps, photofloods
4100°K ~ Moonlight
5500°K ~ Average Daylight
5500°K ~ Electronic Flash (varies)
5770°K ~ Effective sun temperature
6420°K ~ Xenon arc lamp
9300°K ~ TV screen (CRT)

Light Colour

- Temperatures
  - **Example:** if we wanted to light a scene (real or virtual) and we wanted to bathe the main character in moonlight, we would have to set the colour temperature to 4100°K
  - For the same scene, but where we are replicating the effect of using a candle, the colour temperature would have to drop to 1850°K for it to match

Light Colour

- Temperatures
  - Colour Temperature is sometimes used to approximately indicate the "**white balance**" or "**white point**"
  - However, this is a slight misnomer as colour temperature has only one degree of freedom, whereas white balance has two (R-Y and B-Y)

Light Colour

- Uses of Colour Temperature
  - Traditional Film is made for specific light sources (most commonly daylight film and tungsten film), and if used properly, will create a neutral colour print
  - Matching the colour sensitivity of the film to the colour temperature of the light source is one way to balance colour
  - If tungsten film is used indoors with incandescent lamps, the yellowish-orange light of the tungsten [incandescent] bulbs will appear as white (3200K) in the photograph

Light Colour

- Uses of Colour Temperature
  - Filters on a camera lens, or colour gels over the light source(s) may also be used to correct colour balance
  - When shooting with a bluish light (high temperature) source such as on an overcast day, in the shade, in window light or if using tungsten film with white or blue light, a yellowish-orange filter will correct this
  - For shooting with daylight film (calibrated to 5600 K) under warmer (low temperature) light sources such as sunsets, candle light or tungsten lighting, a bluish filter may be used

Light Colour

- Uses of Colour Temperature
  - Experimentation with colour temperature is obvious in many films
    - **Example:** Stanley Kubrick used to, in films such as Eyes Wide Shut, have the light coming in from a window almost always conspicuously blue, whereas the light from lamps on end tables was fairly orange
  - Indoor lights typically give off a yellow hue; fluorescent and natural lighting tends to be more blue

Light Colour

- Uses of Colour Temperature
  - Video camera operators can also white-balance objects which are not white, downplaying the colour of the object used for white-balancing
    - **Example:** we can bring more warmth into a picture by white-balancing off something light blue, such as faded blue denim; in this way white-balancing can serve in place of a filter or lighting gel when those are not available

Light Colour

- Uses of Colour Temperature
  - Cinematographers do not “white balance” in the same way as video camera operators: they can use techniques such as filters, choice of film stock, pre-flashing, and after shooting, colour grading (both by exposure at the labs, and also digitally, where digital film processes are used)

Light Colour

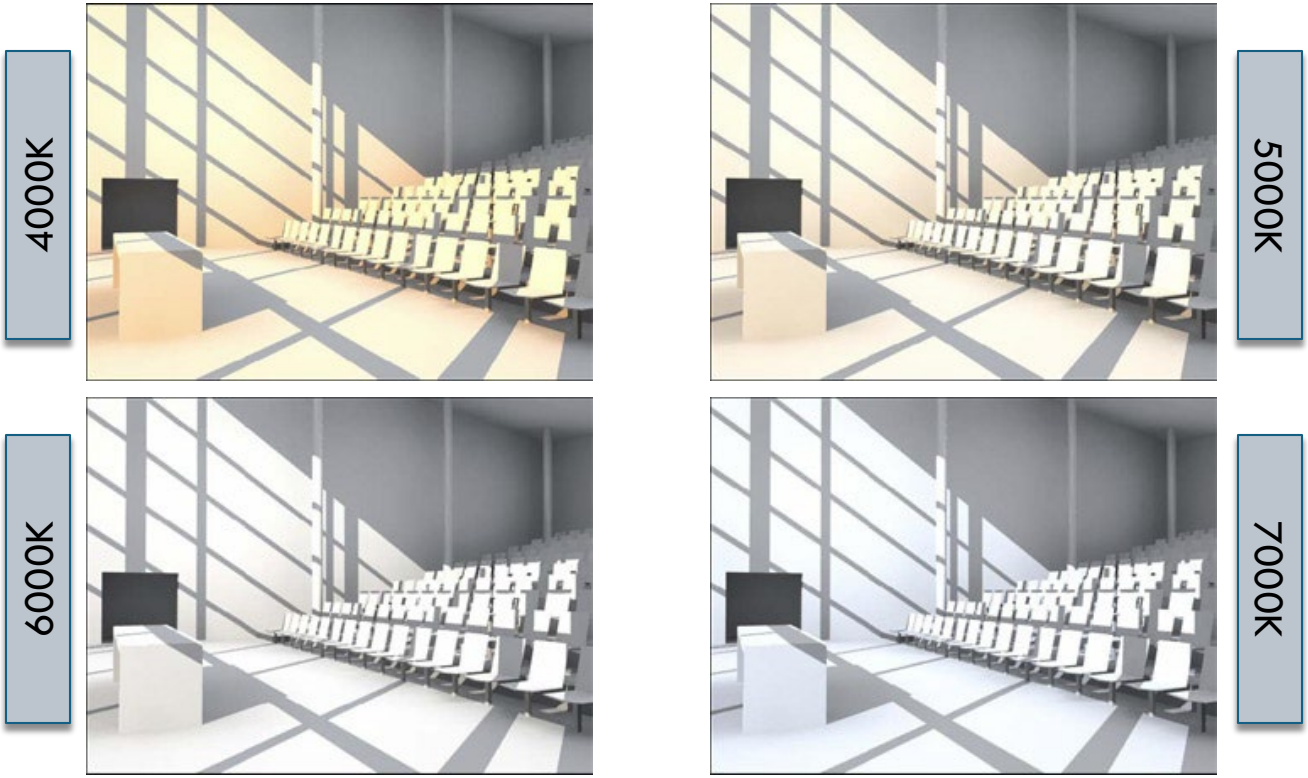
- Uses of Colour Temperature
  - When using a specific light (whether it is a key light, fill light, or back light) the colour temperature of the bulb can be specified
  - Therefore, different bulbs create different light colours – however most key light bulbs are around 3200°K

Light Colour

- Colour Rendering Index
  - A Colour Rendering Index, CRI (sometimes written $\mathbf{R_a}$) is often quoted to indicate how accurately that lamp will portray colours relative to a blackbody source at the same nominal colour temperature
  - All blackbody sources have a CRI of 100
  - CRI values of 80-85 are classified by the manufacturers as 'good' or 'very good' colour-rendering
  - **Example:**
    - Fluorescent lamps have CRI values in the range 55-85

Light Colour

- Virtual Lights
  - As previously mentioned, lighting can be setup in any decent rendering system to replicate a specific colour temperature
  - The default setting for a light, or a basic light, often ignores the colour temperature, and it is often a special light provided by the rendering package (e.g. Arnold etc); this mainly because the colour temperature is a heavier calculation

Light Colour

- Virtual Lights
  - **Example:** images rendered by Mental Ray at different colour temperatures